

$$\begin{aligned}
x_t &= \frac{a}{x_{t-1} + b} \\
x &\rightarrow \alpha x' \\
\alpha x'_t &= \frac{a}{\alpha x'_{t-1} + b} = \\
&= \frac{a}{b(\frac{\alpha}{b} x'_{t-1} + 1)} = \\
&= \frac{a}{b} \frac{1}{\frac{\alpha}{b} x'_{t-1} + 1} \\
\alpha &= b \\
bx'_t &= \frac{a}{b} \frac{1}{x'_{t-1} + 1} \\
x'_t &= \frac{a}{b^2} \frac{1}{x'_{t-1} + 1} = \\
&\stackrel{\frac{a}{b^2}=c}{=} \frac{c}{x'_{t-1} + 1}
\end{aligned} \tag{1}$$